

VolumetricSynth Midterm Presentation

progress update

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Spring 2026



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Introduction

Project Introduction

- VolumetricSynth is a JUCE-based plugin for real-time 3D timbre blending.
- It extends classic 2D vector synthesis to an 8-corner 3D interpolation model.
- Goal: improve expressive sound design while keeping control intuitive.
- This presentation: goals, ownership, progress, remaining work, and timeline to May 5.



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Goals and Scope

- **Goal 1:** Build stable core DSP blocks for a volumetric synthesis pipeline.
- **Goal 2:** Integrate modules into a playable plugin path (MIDI to audio out).
- **Goal 3:** Provide a clear UI/control layer for 3D mix behavior.
- **In scope:** oscillator/filter/envelope/mixer integration, plugin wiring, tests, and core UI state sharing.
- **Out of scope:** advanced FX suite, large preset ecosystem, and deep visual polish.

Module Overview

- DSP/: DualFilter, TrilinearMixer8, oscillation and envelope logic.
- Plugin/: processor/editor lifecycle and parameter plumbing.
- UI/: mixer gain views and controls tied to DSP state.
- Tests/: unit coverage for DualFilter, TrilinearMixer8, and math utilities.
- **Flow:** MIDI/Input → Voice/DSP path → Mixer/Filter/Envelope → Output + UI feedback.

Block Diagram Placeholder

and place it here.

Individual Responsibilities

- **Architecture/Integration (Role):** plugin skeleton, module boundaries, build integration, and validation/QA ownership.
- **Audio Engine (Role):** voice management, routing, and end-to-end realtime-safe orchestration.
- **DSP / Audio Effects (Role):** DSP building blocks (e.g., mixer, filters), algorithm implementation and tuning.
- **UI + State Handoff (Role):** editor controls, gain visualization, and parameter linking.

Progress So Far (Since Jan 20)

- Core repository and CMake structure established.
- Foundational DSP modules implemented (DualFilter, TrilinearMixer8, math helpers).
- Plugin processor/editor path in place for iterative integration.
- Unit tests added for critical DSP/math behavior.
- Current status: strong module-level progress; full end-to-end polish still in progress.

Remaining Tasks and Risks

- Finish full voice/engine integration path and verify stability under load.
- Tighten parameter mapping and edge-case behavior between UI and DSP.
- Expand testing for integration-level scenarios (not only unit modules).
- **Risks:** integration regressions, time pressure near deadline, and performance surprises.
- **Mitigation:** freeze dates, incremental integration checkpoints, and a buffer window before submission.

Comprehensive Timeline (Jan 20 to May 5)

Time	Focus	Owners	Risk/Dep	Done
Jan 20–Feb 2	Kickoff + architecture	A + D	Scope unclear	Build baseline OK
Feb 3–Feb 23	DSP core modules	B	Complexity	Module tests pass
Feb 24–Mar 16	Plugin integration	A + C	API mismatch	Audio path works
Mar 17–Mar 31	UI + state + tests	C + D	Sync issues	UI updates reliably
Apr 1–Apr 14	Feature completion (1)	B + C	Scope creep	Core features done
Apr 15–Apr 24	Hardening (2)	All	Hidden regressions	Stable integrated build
Apr 25–May 1	Demo + docs + rehearsal	All	Late breakages	Script + slides ready
May 2–May 5	Buffer + polish + submit	All	Deadline pressure	Submission package delivered

Demo

- video demo
- thanks, questions?

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